DO NOT PUT ANYTHING HERE

Multi-Subject SISO

DO NOT PUT ANYTHING HERE

Bhavesh, Gorkem, Aditya

DO NOT PUT ANYTHING HERE

Motivation & Introduction

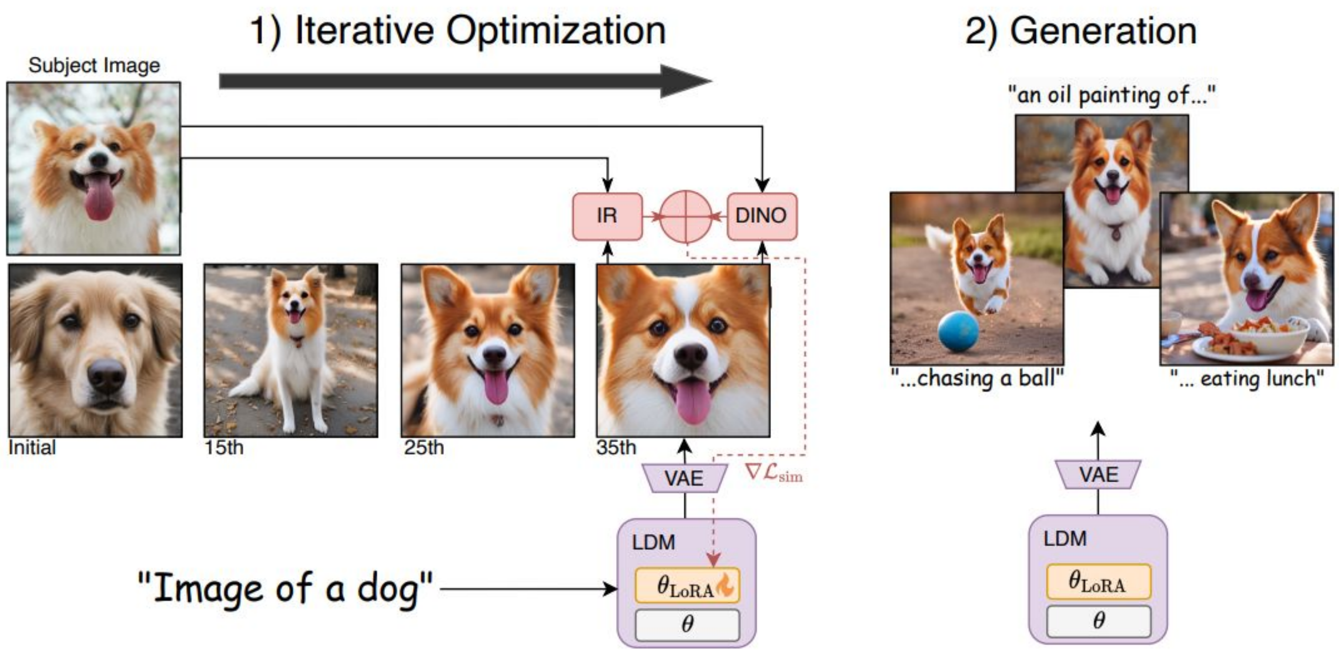
- **Subject-driven generation**: Personalize models with a single subject image.
  - Provide a subject image + a prompt describing an action → generate a new image of the subject doing that action.
- **Current focus**: Only **one** subject per generation (ex: DreamBooth, SISO).
- **Real-world need**: **Multiple subjects** in the same image (e.g., "cat and dog playing").
- **Challenge**: Preserve **identity** and **quality** for each subject.**Goal**: Extend SISO to **multi-subject personalization** with minimal quality loss.

An image of your cat

Prompt: An image of cat as a Sith lord with red lightsaber

Prompt: An image of cat at the space

SISO



1 Sample fixed noise `z_T`

- Like usual diffusion, we start from a Gaussian noise in latent space.

2 Denoise using current model with LoRA

- Apply the U-Net with LoRA-injected cross-attention.
- Condition on text prompt.

3 Decode latent to image via VAE

- Get image `x^_i`.

4 Compare `x^_i` to reference image `x_s` using DINO and IR:

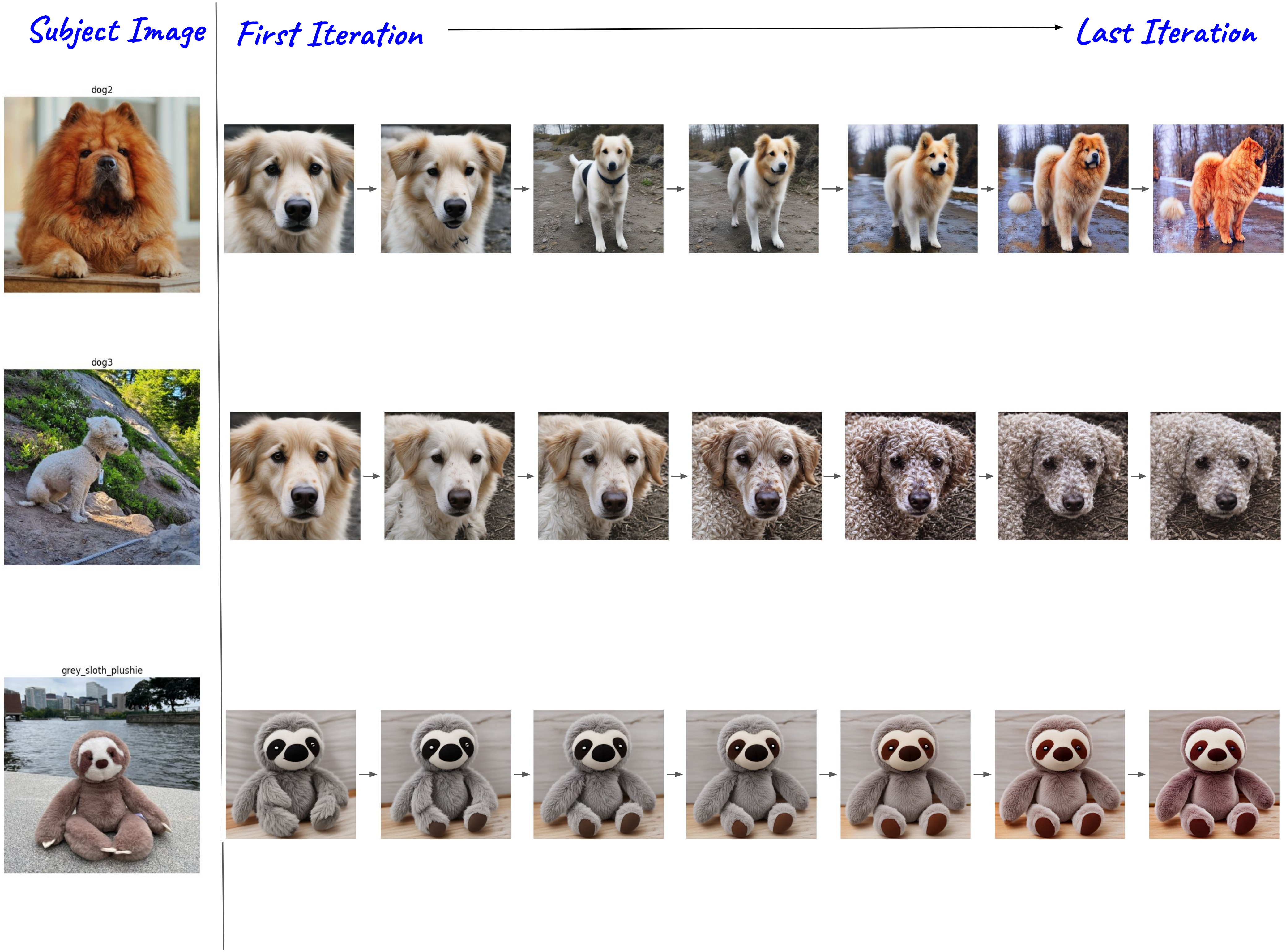
$$\mathcal{L}_{\text{sim}}(\hat{x}_i, x_s) = a \cdot \delta_{\text{DINO}}(\hat{x}_i, x_s) + b \cdot \delta_{\text{IR}}(\hat{x}_i, x_s),$$

5 Backpropagate `L_sim` through U-Net to update LoRA only

- LoRA learns how to shift attention during denoising so that **subject identity is captured.**

6 Once fine-tuned on the Subject Image, generate a new image

- Give more complex prompts
- Can use additional denoising steps for final generation


Subject Image
First Iteration
Last Iteration
dog2
dog3
grey_sloth_plushie

Multi-Subject SISO Masked Algorithm

- **Idea**: Apply **subject-specific masks** to separate regions in the generated image.
- **How?**:
  - Use **GroundingDINO** for object detection (find subject regions).
  - Use **SAM** (Segment Anything Model) for fine-grained subject masks.
  - For each subject, apply mask, compute DINO+IR loss separately.
- **Loss:**

$$\mathcal{L}_{total} = \mathcal{L}_{sim}(x_{gen}, x_{ref}^{A}, m_A) + \mathcal{L}_{sim}(x_{gen}, x_{ref}^{B}, m_B)$$

- **Goal**: Independently optimize weights for each subject inside the same image.
- **Question - 1: How Masked Algorithm worked for single subject?**

Image of wolf_plushie

One of the training time generated images

Mask of that generated image

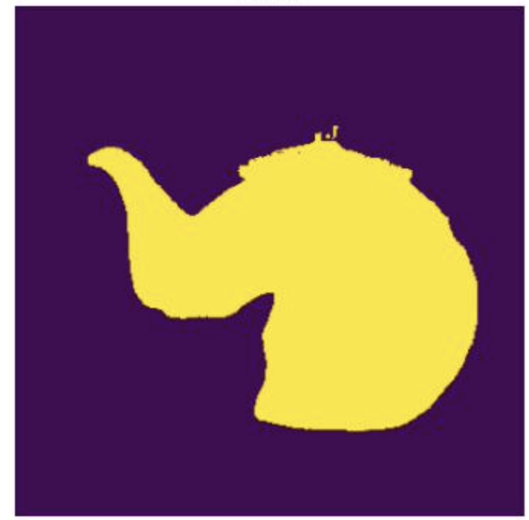
Image of a wolf_plushie with a cup of coffee

Multi-Subject SISO Masked Algorithm

- **Question - 2: How Masked Algorithm worked when there are multiple subjects?**
  - Not quite there, Grounding DINO suffers greatly trying to find a correct bounding box.

Subject Images

Image Generated in the first iteration.



Failed Mask (Both Masks are same)

Prompt: Image of a cat and teapot on a table.

Question - 3: Where is the cat?

- We lost it with masks.

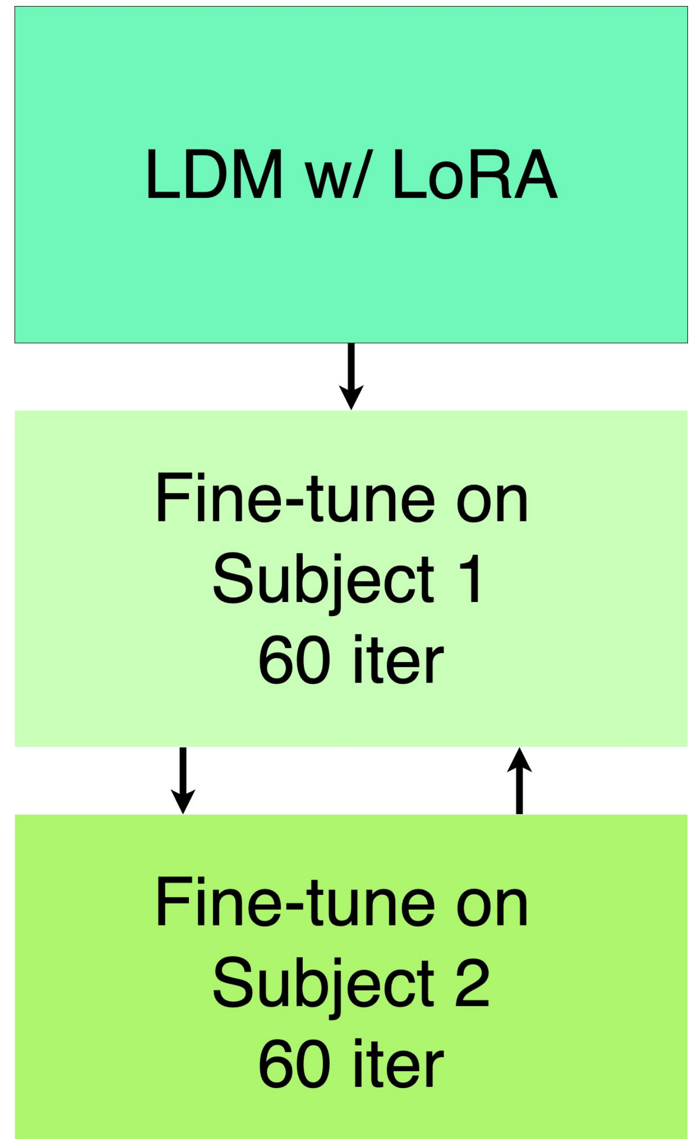
Multi-Subject SISO
Alternate the Training

For each subject:
    Simple prompt for subject
    Until stopping criteria:
        1. Generate image with prompt
        2. Calculate DINO and IR loss
        3. Backpropagate loss

Generate image with more complex prompts with personalized subjects.

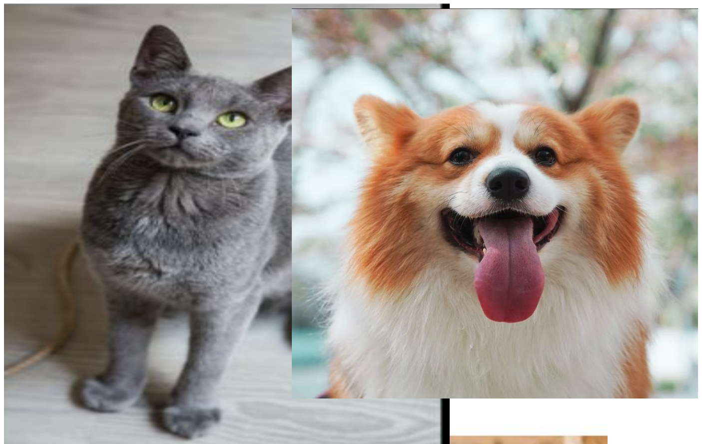
Question: Stop training when?

- Future work and working on it.



Multi-Subject Examples

Reference Photos



Loss

Model	LPIPS	FID	KID
TIGIC	0.22	143.3	0.0066
SwapAnything	0.11	185.7	0.0277
SISO - Single Subject (paper)	0.14	114.8	0.0031
SISO - Multi Subject (ours)	0.41	199.4	0.24

Model	DINO	IR
SISO - Single Subject (paper)	0.48	0.53
SISO - Multi Subject (ours)	0.84	0.89

Expected larger loss due to multi-subject over single-subject.

Limitations

- Object detection models can misclassify and give large bounding box, leading to incorrect masks
- Memory requirements requires model to run on A100
  - Even more than 40 GB GPU memory needed
- Model requires fine-tuning leading to increased latency on first sample generation
- Similar subjects may morph features (ex: cat and tiger)

Future Work:

- For the Multi-Subject Alternate the Training, a better implementation of round-robin fashion training and early quits are needed
- Trying reverse masking for the Multi-Subject Masked Algorithm. Generally one of the subjects is the dominant and all the masks show that subject. But in case of 2 subjects, one might try one mask with reversing the values.